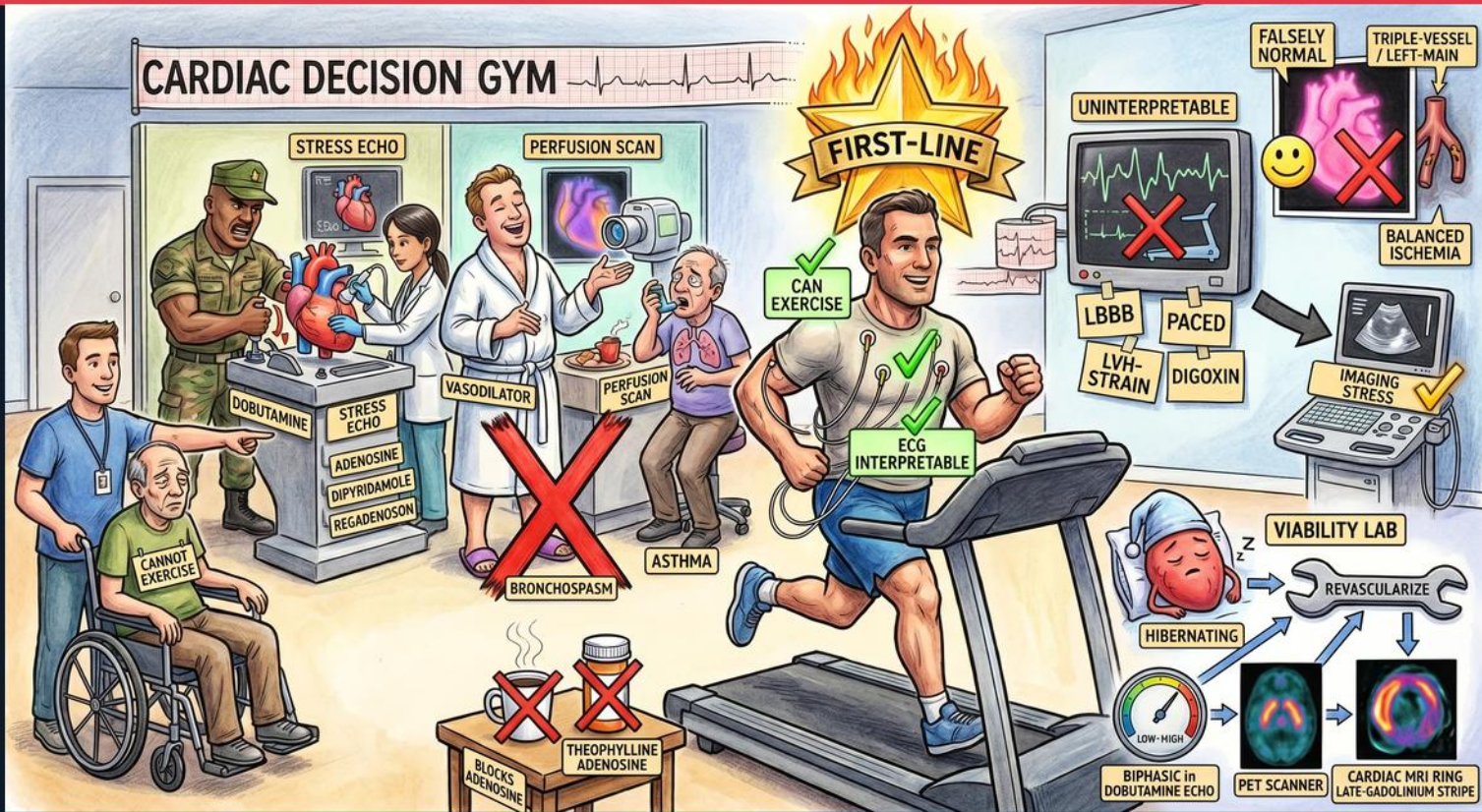


Cardiac Exam — Stress Testing & Viability



15 symbols · sketch walkthrough



FIRST-LINE star

WHAT YOU SEE

Glowing yellow 'FIRST-LINE' star/medal above the treadmill.

WHAT IT ENCODES

Exercise treadmill ECG is the FIRST-LINE stress test, but only when BOTH conditions hold: the patient CAN physically exercise to an adequate workload AND the baseline ECG is INTERPRETABLE for ST changes. Exercise adds functional and prognostic data that drugs cannot, and an interpretable ECG makes ST analysis diagnostic. If either condition fails, move to a pharmacologic and/or imaging stress test.

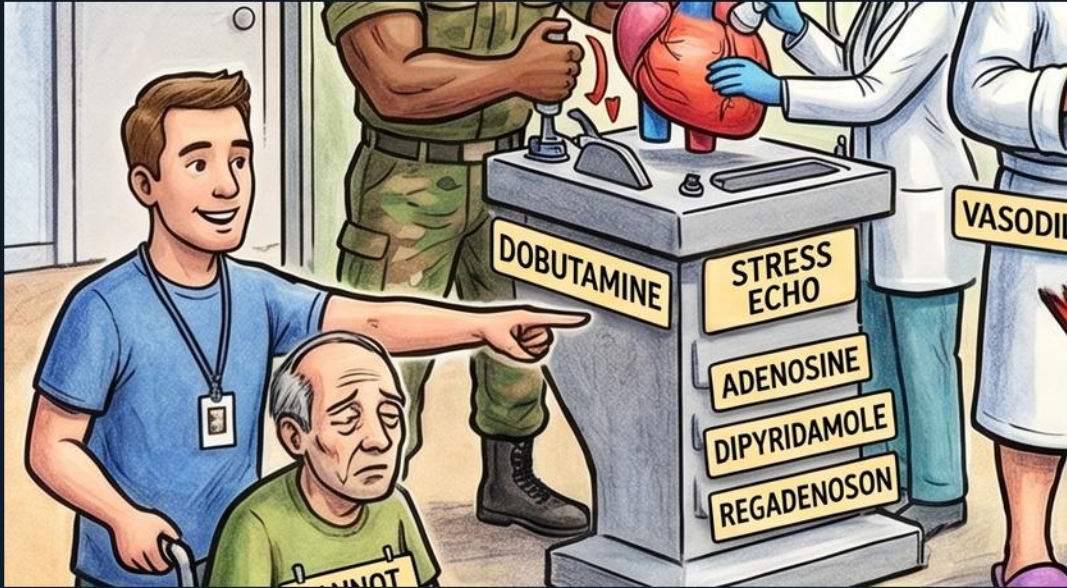
Cannot exercise patient

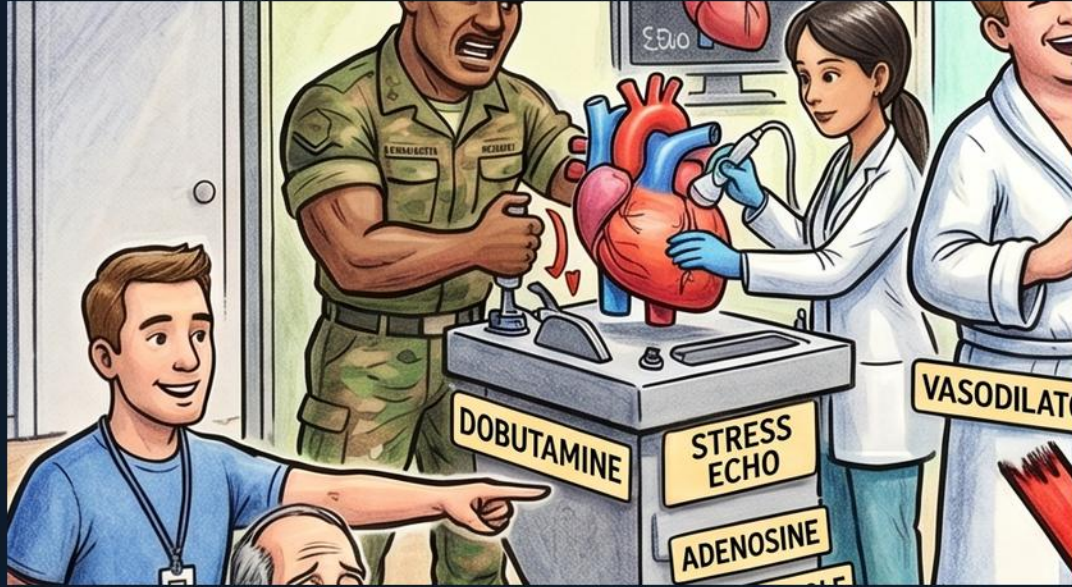
WHAT YOU SEE

Elderly man in a WHEELCHAIR labeled 'CANNOT EXERCISE' at far left.

WHAT IT ENCODES

If the patient CANNOT exercise (e.g., frailty, arthritis, PAD, deconditioning), switch to a PHARMACOLOGIC stress test. The choice is then between an inotrope (dobutamine stress echo) and a vasodilator (perfusion imaging). Inability to exercise, not the ECG, drives this branch.





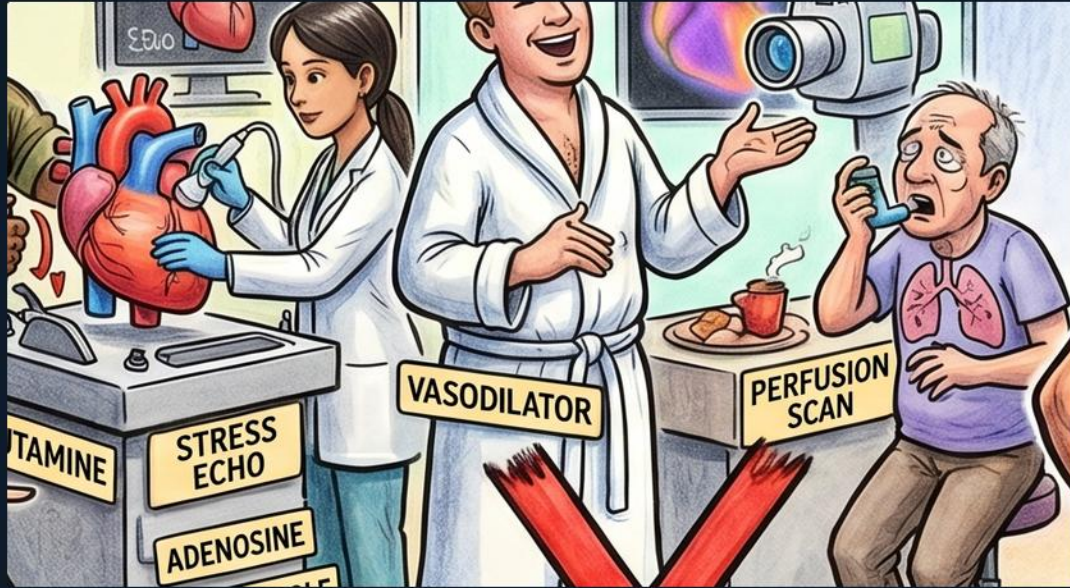
Dobutamine stress echo

WHAT YOU SEE

Drill-sergeant beating a heart, panel 'DOBUTAMINE / STRESS ECHO'.

WHAT IT ENCODES

Dobutamine is an INOTROPE that increases heart rate and contractility to provoke ischemia for stress ECHOCARDIOGRAPHY. It is a pharmacologic option for patients who cannot exercise. New or worsening wall-motion abnormalities indicate inducible ischemia.



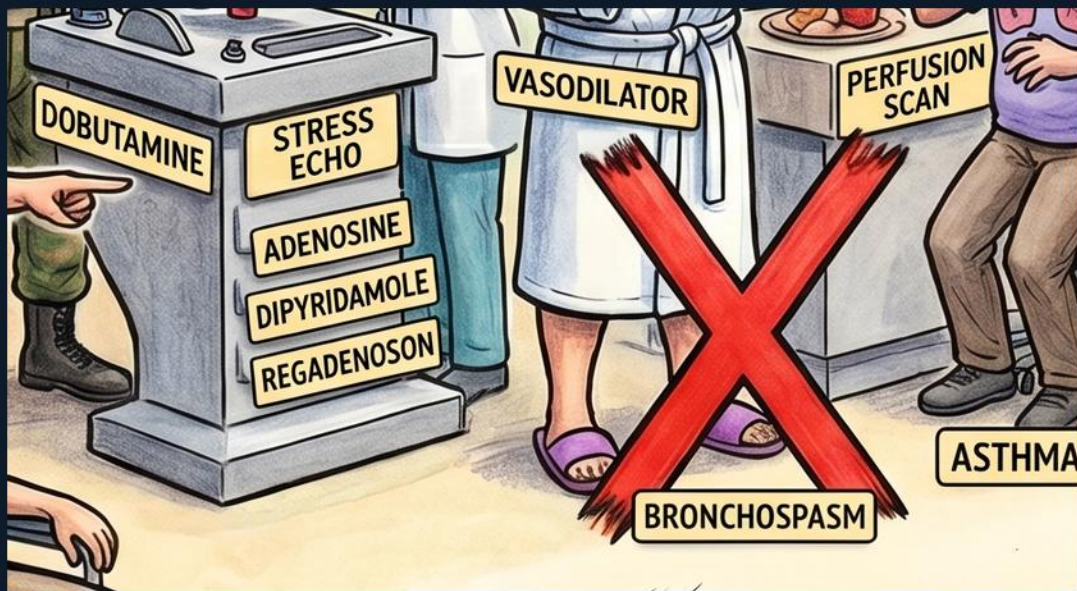
Vasodilator perfusion panel

WHAT YOU SEE

Panel 'VASODILATOR — ADENOSINE / DIPYRIDAMOLE / REGADENOSON' beside a perfusion camera.

WHAT IT ENCODES

Vasodilators (adenosine, dipyridamole, regadenoson) are used for PERFUSION imaging (nuclear MPI or stress perfusion). They dilate normal coronaries and create heterogeneity of flow that unmasks stenosed territories. They are the pharmacologic partner to perfusion imaging when the patient cannot exercise.



Bronchospasm — crossed out

WHAT YOU SEE

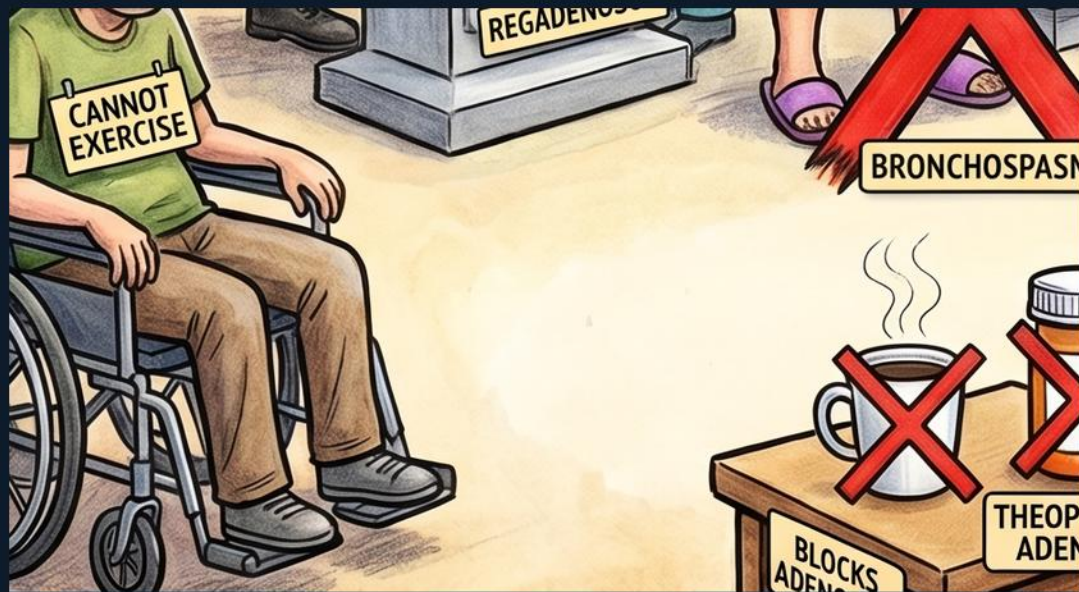
Red X over a panel labeled 'BRONCHOSPASM' / 'ASTHMA'.

WHAT IT ENCODES

AVOID vasodilator stress agents (adenosine, dipyridamole, regadenoson) in patients with active bronchospasm or asthma, because they can precipitate bronchoconstriction. In such patients use dobutamine stress echo instead. This is a key safety contraindication.

THE BOARD TRAP

Giving adenosine/dipyridamole to a bronchospastic/asthmatic patient is dangerous — choose dobutamine for these patients.



Theophylline blocks adenosine

WHAT YOU SEE

Coffee cup and pill bottle labeled 'THEOPHYLLINE' / 'BLOCKS ADENOSINE' with a red X.

WHAT IT ENCODES

CAFFEINE and methylxanthines (theophylline) are adenosine-receptor antagonists that BLUNT the response to adenosine/dipyridamole and cause a false-negative perfusion study. Patients must HOLD caffeine and theophylline (typically 12–24h) before vasodilator stress testing. Failure to do so invalidates the test.

THE BOARD TRAP

Not holding caffeine/theophylline before adenosine or dipyridamole stress blunts the vasodilator and can yield a falsely normal scan.



Uninterpretable ECG sign

WHAT YOU SEE

Monitor labeled 'UNINTERPRETABLE' showing an abnormal ECG with a red X.

WHAT IT ENCODES

If the baseline ECG is UNINTERPRETABLE, ST-segment analysis from a plain treadmill ECG is unreliable. These patients need an IMAGING-based stress test (stress echo or nuclear perfusion) that does not rely on ST changes. The interpretability of the ECG, not exercise ability, drives this branch.



LBBB / paced / LVH / digoxin

WHAT YOU SEE

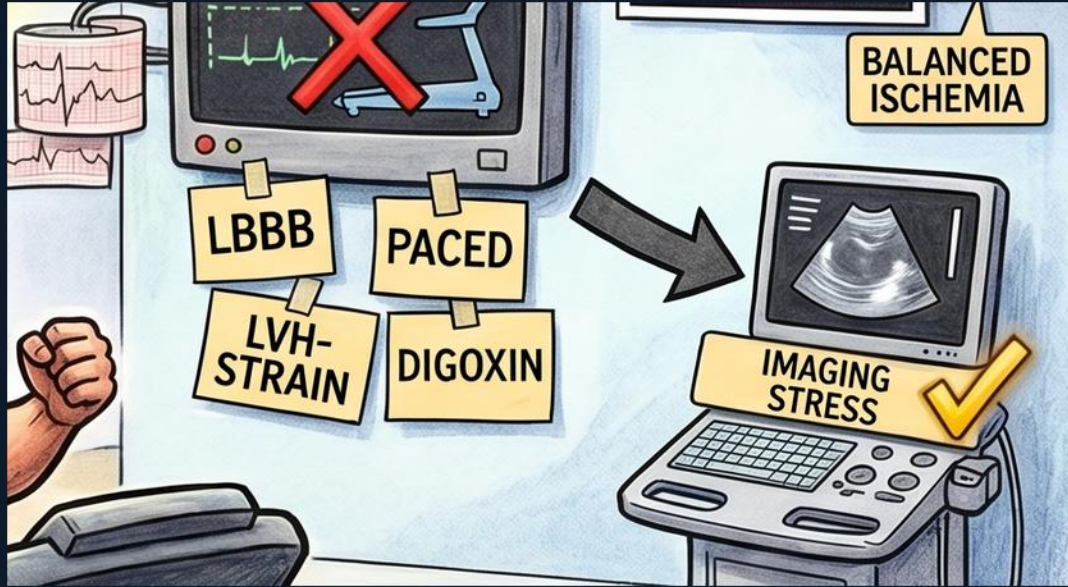
Labels 'LBBB', 'PACED', 'LVH-STRAIN', 'DIGOXIN' under the uninterpretable monitor.

WHAT IT ENCODES

Causes of an uninterpretable baseline ECG include LBBB, ventricular PACING, LVH with strain, pre-excitation, and DIGOXIN effect. In all of these, exercise-induced ST changes cannot be trusted. These patients require imaging stress (echo or nuclear) rather than plain treadmill ECG.

THE BOARD TRAP

Sending an LBBB/paced/LVH-strain/digoxin patient for a plain treadmill ECG is wrong — the ST segments are uninterpretable, so use imaging stress.



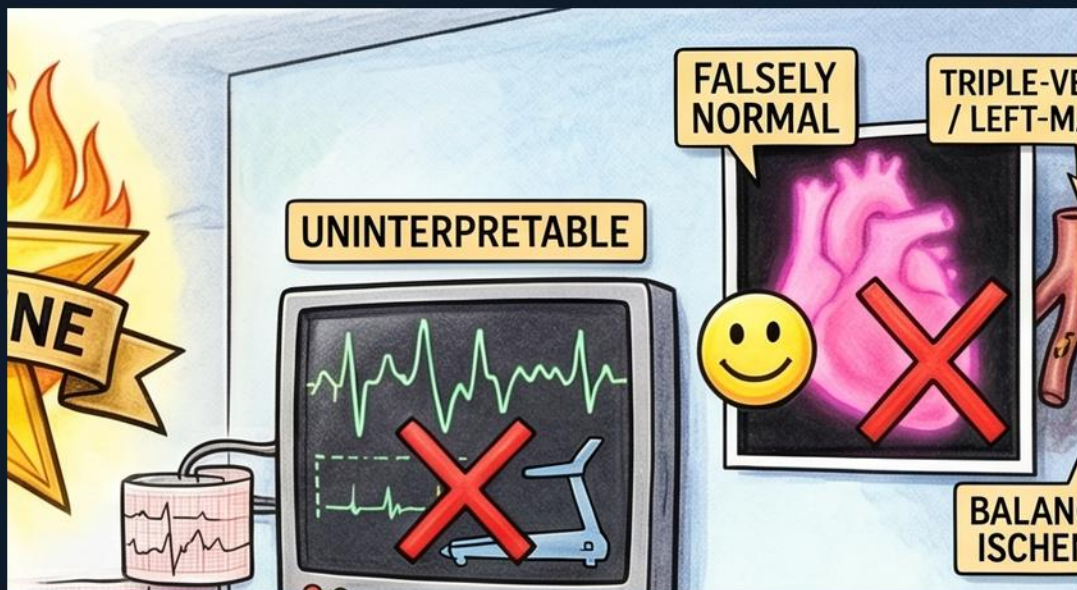
Imaging stress echo monitor

WHAT YOU SEE

Ultrasound machine at right labeled 'IMAGING STRESS'.

WHAT IT ENCODES

IMAGING stress (stress echocardiography or nuclear MPI) is the answer when the ECG is uninterpretable or when localizing ischemia is needed. Imaging detects wall-motion or perfusion abnormalities independent of baseline ST changes. It can be combined with exercise or with pharmacologic stress.



Balanced ischemia — falsely normal

WHAT YOU SEE

Smiley face 'FALSELY NORMAL' with a red X, next to 'TRIPLE-VESSEL / LEFT-MAIN' and 'BALANCED ISCHEMIA'.

WHAT IT ENCODES

In balanced ischemia from THREE-VESSEL or LEFT-MAIN disease, perfusion is reduced UNIFORMLY across all territories, so a RELATIVE perfusion scan can look FALSELY NORMAL. Clues to severe disease include transient ischemic dilation, a drop in ejection fraction with stress, and high-risk clinical features. Recognizing this pitfall prevents falsely reassuring a high-risk patient.

THE BOARD TRAP

A 'normal' relative perfusion scan can hide balanced ischemia in left-main/3-vessel disease — do not call it reassuring without considering global ischemic markers.



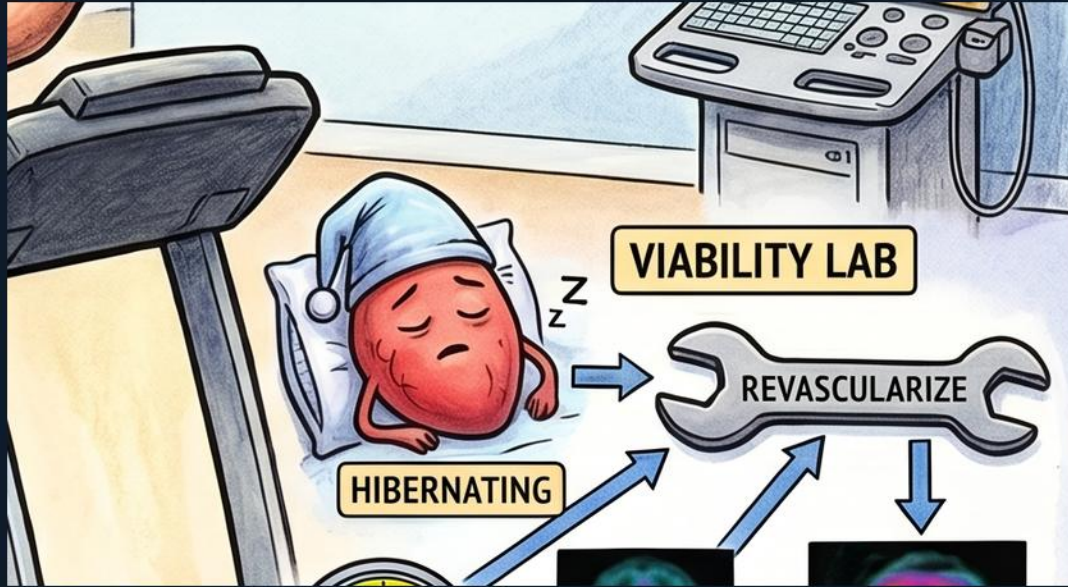
Hibernating myocardium

WHAT YOU SEE

Sleeping heart with 'Z's labeled 'HIBERNATING' in the Viability Lab.

WHAT IT ENCODES

Viability testing identifies HIBERNATING myocardium: chronically hypoperfused but still-living tissue with reversible dysfunction. Such territory recovers function after REVASCULARIZATION, so distinguishing it from scar guides whether to revascularize. Viable, hibernating myocardium is the target worth opening up.



Revascularize arrow

WHAT YOU SEE

Arrow labeled 'REVASCULARIZE' pointing from the hibernating heart.

WHAT IT ENCODES

Demonstrating viable (hibernating) myocardium supports REVASCULARIZATION, because function and prognosis can improve. Conversely, predominantly scarred, non-viable territory is unlikely to benefit. Viability assessment thus selects which patients gain from revascularization.



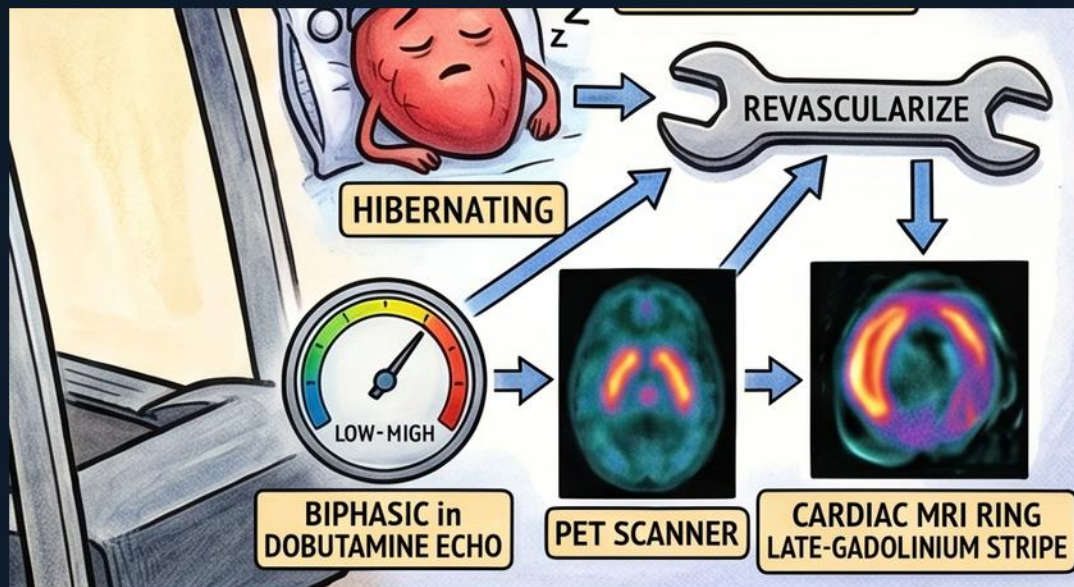
Biphasic dobutamine echo

WHAT YOU SEE

Small gauge labeled 'BIPHASIC in DOBUTAMINE ECHO' (low→high dose).

WHAT IT ENCODES

A BIPHASIC response on dobutamine stress echo — improved wall motion at LOW dose then worsening at HIGH dose — is the classic marker of viable, hibernating myocardium. The low-dose augmentation shows contractile reserve; the high-dose decline shows underlying ischemia. This pattern predicts functional recovery after revascularization.



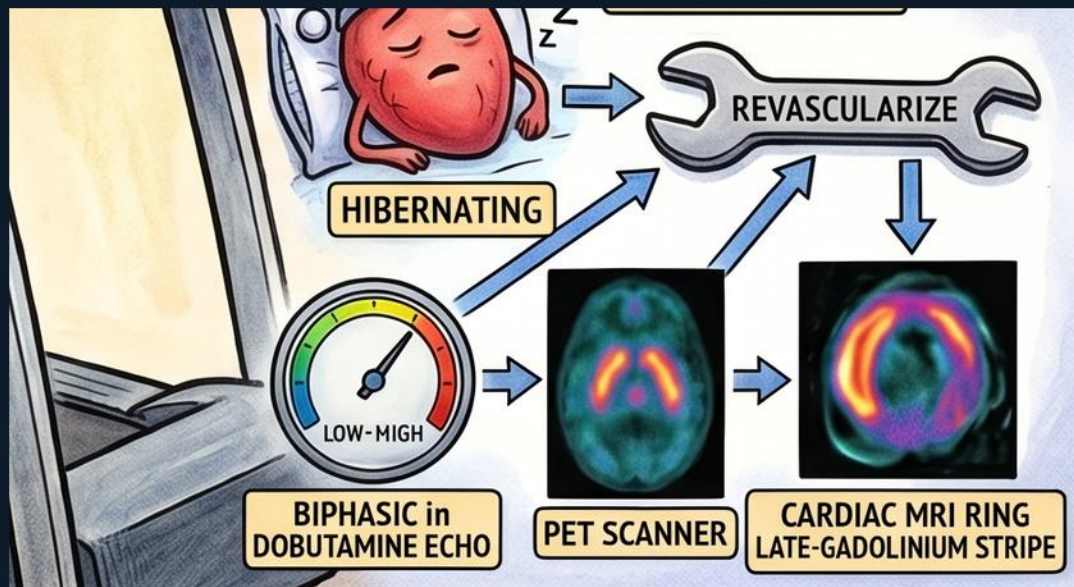
PET scanner

WHAT YOU SEE

PET scanner icon in the Viability Lab.

WHAT IT ENCODES

PET (metabolic/perfusion mismatch) is a viability tool: preserved metabolism with reduced perfusion indicates hibernating, viable myocardium. PET demonstrates tissue that is alive but underperfused and therefore salvageable. It is one of the reference modalities for viability.



Cardiac MRI late gadolinium

WHAT YOU SEE

Heart with a bright 'late-gadolinium' ring, labeled 'CARDIAC MRI LATE-GADOLINIUM'.

WHAT IT ENCODES

Cardiac MRI with LATE GADOLINIUM enhancement maps scar: the transmural EXTENT of enhancement predicts recovery. Little or no late enhancement means viable myocardium likely to recover after revascularization, whereas near-transmural scar will not. LGE-CMR is a leading viability test.

Summary — every symbol

1. FIRST-LINE star

Exercise treadmill ECG is the FIRST-LINE stress test, but only when BOTH conditions hold: the patient CAN physically exercise to an adequate workload AND the baseline ECG is INTERPRETABLE for ST changes

3. Dobutamine stress echo

Dobutamine is an INOTROPE that increases heart rate and contractility to provoke ischemia for stress ECHOCARDIOGRAPHY

5. Bronchospasm — crossed out

AVOID vasodilator stress agents (adenosine, dipyridamole, regadenoson) in patients with active bronchospasm or asthma, because they can precipitate bronchoconstriction

7. Uninterpretable ECG sign

If the baseline ECG is UNINTERPRETABLE, ST-segment analysis from a plain treadmill ECG is unreliable

9. Imaging stress echo monitor

IMAGING stress (stress echocardiography or nuclear MPI) is the answer when the ECG is uninterpretable or when localizing ischemia is needed

11. Hibernating myocardium

Viability testing identifies HIBERNATING myocardium: chronically hypoperfused but still-living tissue with reversible dysfunction

13. Biphasic dobutamine echo

A BIPHASIC response on dobutamine stress echo — improved wall motion at LOW dose then worsening at HIGH dose — is the classic marker of viable, hibernating myocardium

15. Cardiac MRI late gadolinium

Cardiac MRI with LATE GADOLINIUM enhancement maps scar: the transmural EXTENT of enhancement predicts recovery

2. Cannot exercise patient

If the patient CANNOT exercise (e.g., frailty, arthritis, PAD, deconditioning), switch to a PHARMACOLOGIC stress test

4. Vasodilator perfusion panel

Vasodilators (adenosine, dipyridamole, regadenoson) are used for PERFUSION imaging (nuclear MPI or stress perfusion)

6. Theophylline blocks adenosine

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8. LBBB / paced / LVH / digoxin

Causes of an uninterpretable baseline ECG include LBBB, ventricular PACING, LVH with strain, pre-excitation, and DIGOXIN effect

10. Balanced ischemia — falsely normal

In balanced ischemia from THREE-VESSEL or LEFT-MAIN disease, perfusion is reduced UNIFORMLY across all territories, so a RELATIVE perfusion scan can look FALSELY NORMAL

12. Revascularize arrow

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14. PET scanner

PET (metabolic/perfusion mismatch) is a viability tool: preserved metabolism with reduced perfusion indicates hibernating, viable myocardium